Introduction and EDA

Development Team Project: Project Report

Team F

We also have to send the peer view, each of us (second part of assignment of U6)

Introduction

Airbnb provides short-term accommodation across many locations and price ranges in New York City. Location is likely to be an important factor in Airbnb pricing (SOURCE). Therefore, we assume that distance from the most centrally located touristic area (Times Square; SOURCE) is associated with listing price. To test this hypothesis, we use the 2019 New York City Airbnb dataset. We use regression and clustering to identify different types of Airbnb listings based on location and property characteristics. Thus, our research question is as follows:

How is distance from Times Square associated with Airbnb listing prices in New York City, and what listing segments can be identified based on location and property characteristics?

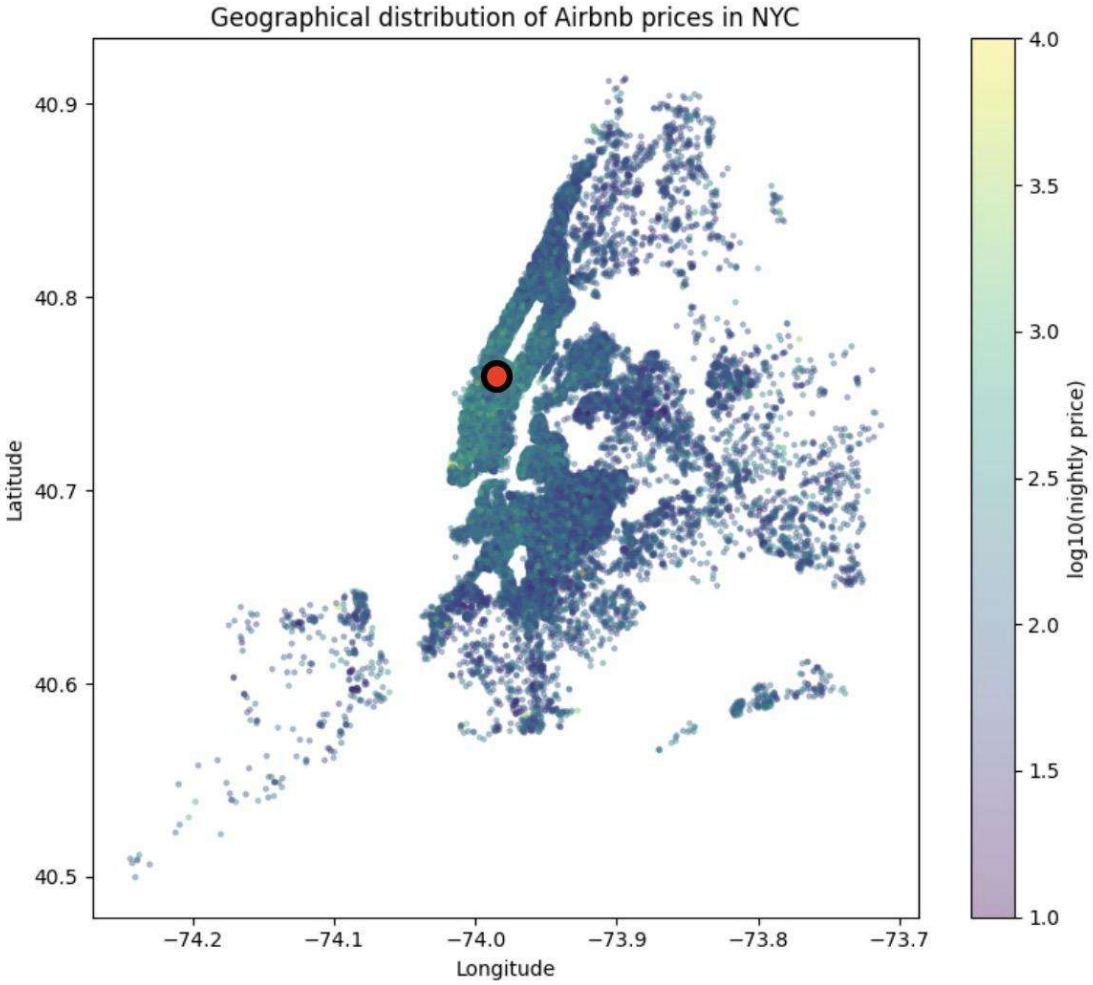
The main purpose of this research is to generate insights that could help Airbnb understand pricing patterns and different listing segments in New York City.

Data preparation

We first cleaned the original dataset that had 48,895 Airbnb listings. If there were no values for reviews_per_month, it was changed to 0. Unnecessary identifier columns such ad id, name, host_id, and host_name were removed. We deliberately kept extremely high prices since there was no evidence that these were mistakes. We did remove 11 listings with a price of $0. Thus, the final cleaned dataset consisted of 48,884 listings. As a final step, we created the distance_times_square_km column which calculates distance from Times Square using latitude and longitude.

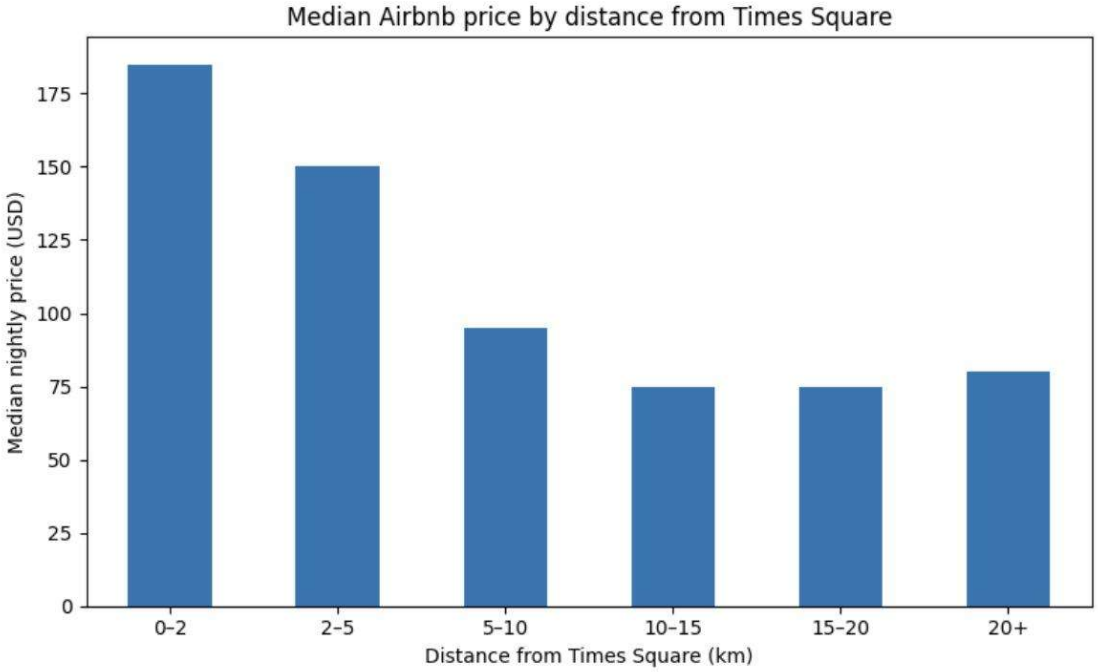
Exploratory Data Analysis

We first visualized listing prices according to geographic location. This resulted in the following map:



The red dot represents the location of Times Square. We can clearly see that higher prices are concentrated around Manhattan/Times Square. Listings further away are generally cheaper.

Next, we visualized the median Airbnb price by distance from Times Square:



Listings within 0-2 km of Times Square have a median price of $185. At 2-5 km, this falls to $150. At 5-10 km, it falls even lower to $95. If the property is further away than 10 km, the price is around $75.

The pattern, therefore, is that Airbnb prices tend to decrease as distance from Times Square increases. Since the price slightly increases again from 20+ km, the relationship is not perfectly straight.

However, distance from Times Square might not be the only reason prices gradually decrease. Listings closer to Times Square may also be different because of borough, room type, quality, neighborhood and other characteristics. Therefore, regression and clustering are used to investigate these relationships in more detail.

References:

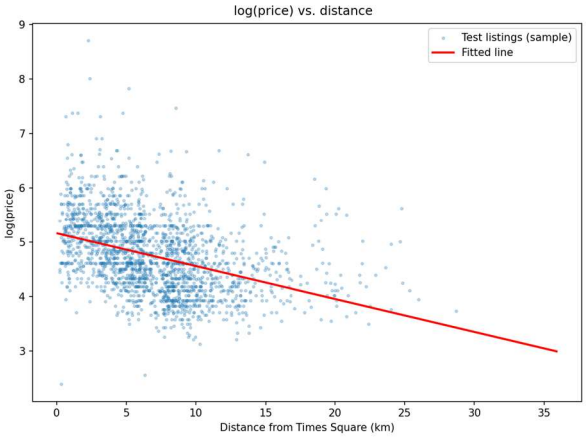
ChatGPT conversation on data cleaning and Exploratory Data Analysis:
https://chatgpt.com/share/6a88935e-4bc8-83eb-9413-8584ef5f1ff3

Regression

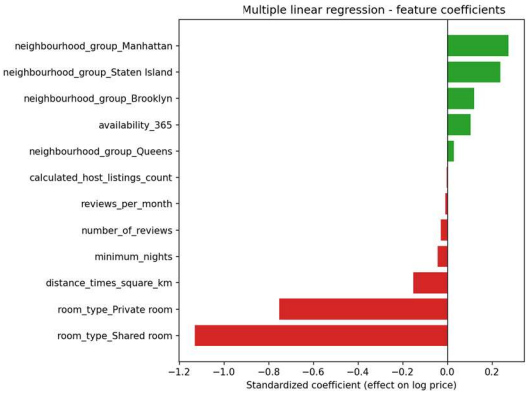
Framing our question as a regression problem means finding a model that predicts the value of price based on how one or more variables change. To do so, we firstly decided to transform price into its logarithmic form log(price) to reduce the impact of the few listings that have an outlying price (5-10k £).
Then we checked multicollinearity [**REF whats colinearity**] amongst distance and borough to see whether it would be difficult for the model to separate their individual effects on price: distance was not problematic (VIF = 1.9, Variance Inflation Factor) [**REF how to interpret vif**], borough dummies though showed a VIF >10, which is high. This warned us that the split of credit between borough and distance should not be taken as separate causal effects.
Secondly, the data was split into a test and a training set, respectively containing 20% and 80% of the data. To do so we also stratified by borough to avoid underrepresenting the rarest labels [**REF on stratified split**].

We then used three different regression approaches and compared them:

1. A simple linear regression [**REF** linear reg] on distance alone found each +1km was associated with a ~5.9% price drop (R^2=0.15).

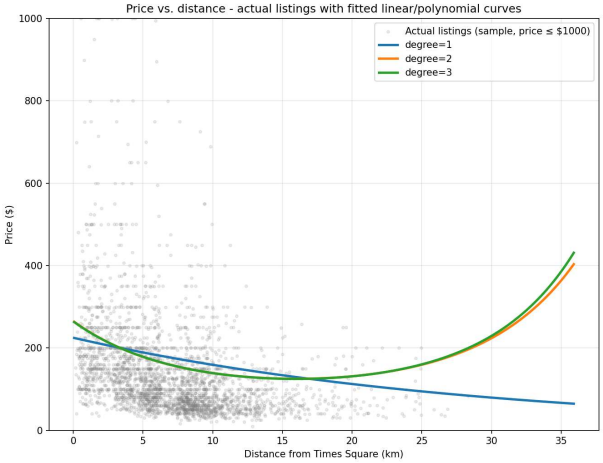


2. A multilinear regression [**REF** multilin reg] approach with all features; controlling for room type and borough caused distance's effect to shrink, showing that part of its raw association in the simple linear model was probably an effect mediated by borough.



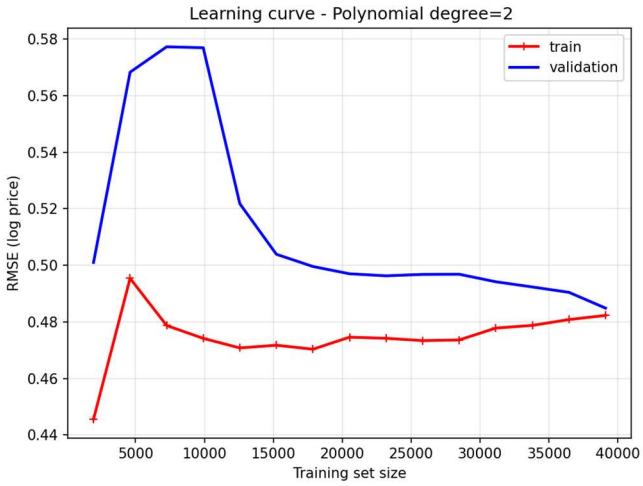
3. Polynomial regression [**REF** polynom reg]: we checked first with degree 2, which improved fit (R^2=0.51), revealing price drops steeply near the center and flattens further out. Degree 3 didn't add any significant improvement. In the figure, degree 1

is the linear model.



The degree-2 polynomial achieved a test RMSE of 0.488 (log-price units) and R^2 of 0.512, both an improvement over the plain linear model (RMSE 0.496, R^2 0.496). This confirms that the true relationship between distance and price is not linear.

Ultimately, the learning curves showed train/validation RMSE converging closely, which told us that the model fairly predicts price, without overfitting the data.



Because there were no signs of overfitting, we chose not to use any regularization technique on the model.

Clustering

While regression examined the relationship between listing price, distance and other characteristics, clustering was used to identify distinct groups of Airbnb listings with similar location and property characteristics, providing a complementary perspective on the patterns within the dataset. **[REF: clustering/unsupervised learning]**.

The optimal number of clusters was evaluated using the elbow method **[REF]** and average silhouette scores **[REF]**. Although k = 2 produced the highest silhouette score (0.392), this solution was considered too coarse to provide meaningful segmentation beyond the existing price–distance findings. The three-cluster solution achieved the next highest substantively meaningful silhouette score (0.252) and corresponded with the point at which improvements in within-cluster variation began to flatten; therefore, k = 3 was selected.

K-means clustering **[REF]** was performed on six scaled variables **[REF: standardisation/feature scaling]**. Figure X uses principal component analysis (PCA) **[REF: principal component analysis (PCA)]** to visualise the resulting clusters in two dimensions; clustering was not performed on the principal components themselves. Together, PC1 and PC2 explain 52.5% of the variance. The clusters occupy distinguishable regions of the PCA space, although some overlap remains.

Figure X: PCA projection of K-means Airbnb listing clusters (k = 3)



(Source: Team F analysis of AB_NYC_2019 dataset using R.)

Cluster profiling (Table 1) indicates that Cluster 1 represents a smaller professional/long-stay segment, characterised by a 30-night median minimum stay, high availability (316 days) and hosts with a median of 16 listings. Cluster 2 represents active value short-stay listings, with the lowest median price ($95), greatest distance from Times Square (8.7 km) and highest review activity (2.12 reviews/month). Cluster 3 comprises the largest segment (54.6%), characterised by zero median availability and low review activity.

Table 1: Profile Characteristics of the Three K-Means Airbnb Listing Clusters

	Cluster 1	Cluster 2	Cluster 3
Size	4,594 (9.4%)	17,608 (36.0%)	26,682 (54.6%)
Median price	$150	$95	$115
Median distance from Times Sq.	3.9 km (closest)	8.7 km (furthest)	5.7 km (middle)
Median minimum nights	**30**	2	3
Median # reviews	1	28	2
Median reviews/month	0.09	2.12	0.12
Median availability (days/365)	**316**	197	**0**
Median host listings count	**16**	1	1
Dominant room type	Entire home/apt (68%)	Private room (54%)	Entire home/apt (55%)
Dominant borough	Manhattan (70%)	Brooklyn (44%), spread across boroughs	Manhattan (49%) / Brooklyn (43%)

(Source: Team F analysis of the AB_NYC_2019 dataset using R.)

Overall, cluster prices decline with increasing distance from Times Square ($150, $115 and $95), supporting the broad regression pattern while demonstrating that listing segments also differ substantially in host, availability and booking characteristics. However, the moderate silhouette score and overlap in the PCA projection indicate that these segments should be interpreted as broad listing profiles rather than sharply separated groups.